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HIGHLIGHTS

- A sizable amount of data is collected by analyzing collision reports and news outlets.
- The estimated models offer insights into various variables related to injury severity and vehicles equipped with self-driving technologies.
- A comparison and analysis are made between the factors that impact injury severity involving the ADAS and ADS equipped vehicles.

ABSTRACT

Vehicles equipped with automated driving capabilities have shown the potential to improve safety and operations. Advanced driver assistance systems (ADAS) and automated driving systems (ADS) have been widely developed to support vehicular automation. Although the studies on the injury severity outcomes that involve automated driving systems are ongoing, there is limited research investigating the difference between injury severity outcomes of the ADAS and ADS vehicles using real-world crash data. To ensure comprehensive analysis, a multi-source dataset that includes the NHTSA crash database (752 cases), CA DMV crash reports (498 cases), and news outlet data (30 cases) is used. Two random parameters multinomial logit models with heterogeneity in the means and variances are estimated to gain a better understanding of the variables impacting the crash injury severity outcome for the ADAS (SAE Level 2) and ADS (SAE Levels 3-5) vehicles. We found that while 56 percent of crashes involving ADAS vehicles took place on a highway, 84 percent of crashes involving ADS took place in more urban settings. The model estimation results indicate that the weather indicators, traffic incident/work zone indicator, differences in the system sophistication that are captured by both— manufacture year and high/low mileage, type of collision, as well as rear and front impact

indicators all play a significant role in the crash injury severity. The results offer an exploratory assessment of the safety performance of the ADAS and ADS equipped vehicles in the real-world environment and can be used by the manufacturers and other stakeholder to dictate the direction of their deployment and usage.

Keywords: autonomous vehicle, crash analysis, automated driving systems, advanced driver assistance systems, random parameters multinomial logit model

1. INTRODUCTION

Automation of systems has been undergoing rapid development and has revolutionized the transportation industry. SAE International [1] categorized automated driving systems into six levels. Level 0 and Level 1 features provide limited support in driving and include technologies such as steering or brake support. Level 2 indicates Advanced Driver Assistance Systems (ADAS) and while the driver must still be involved in the vehicle operations, the automation features offer steering and brake support as well as lane centering and adaptive cruise control. Vehicles with automation Levels 3-5 do not need a driver (although for Level 3, a driver must take over if the vehicle software requests it). Levels 3-5 are also referred to as Automated Driving Systems (ADS), and while in some cases their operations will depend on the environment, they offer a high level of automation. Although the advantages of these automated driving systems have been to some degree, already demonstrated in smart transportation, many of these emerging technologies have also brought numerous drawbacks particularly in the context of safety risks.

Around the world, the vehicles equipped with different levels of autonomous driving capabilities have been widely tested on public roads. In California, for example, the manufacturers of the vehicles equipped with automated driving systems, commonly referred to as autonomous vehicles (AVs), are permitted to test their vehicles on public roads by obtaining permits from the California Department of Motor Vehicles (DMV). Although only the approved vehicles and operators are permitted to conduct testing, the manufacturer must identify the specific operational design domain of the autonomous vehicle in the permit application. During testing, the operators must follow the ordinary California traffic regulations, and the manufacturer is required to submit reports to the DMV that include the details of any incident,

such as crash as well as disengagement reports. The disengagement reports are needed when a vehicle equipped with automated driving systems disengages from the automated mode either due to a technology failure or in cases when the operator takes manual control of the vehicle. Disengagement reports also include the total autonomous miles traveled per year (AVMT). On March 17th, 2023, California DMV has authorized the public release of the 564 reports of crashes involving vehicles equipped with automated driving systems (ADS or ADAS) that have occurred since 2014. While the goal of the autonomous vehicles (AVs) and vehicles with varied levels of automation is to create the future with zero road fatalities, the limited testing conducted thus far has resulted in hundreds of crashes in only one state, raising concerns about their safe operations.

To support the timely and transparent release of the records reflecting real-world crashes that involve vehicles equipped with ADS or ADAS from manufacturers and operators, the National Highway Traffic Safety Administration (NHTSA) issued a General Order in June 2021 requiring manufacturers and operators to report all crashes involving vehicles equipped with different levels of driving automation namely Level 2 Advanced Driver Assistance Systems (ADAS) as well as Levels 3 to 5 Automated Driving Systems (ADS). The General Order requires incident reports for crashes that occur on the public roads in the United States. As of May 2022, the NHTSA received incident reports for 130 crashes involving the ADS/ADAS-equipped vehicles. These reports contribute to gaining insights into the safe operations of vehicles equipped with automated driving systems as well as provide crucial data necessary for research and understanding their driving capabilities in diverse environments [2].

In the current paper, we use an integrated data source to conduct exploratory analysis of injury severity involving vehicles equipped with automated driving systems (Level 2 to Level 5). The dataset used for this research combines multiple sources and contains crash reports, weather

data, vehicle manufacturers, and geospatial data. Two random parameters multinomial logit models with heterogeneity in the means and variances of random parameters are estimated to identify factors influencing injury severity for vehicles equipped with ADS and ADAS. The model estimation results provide valuable and novel insights into the risks associated with the evolving autonomous vehicle technology and can be used to improve their safety and further stimulate innovation. The results of this study aim to inform policymakers, engineers, and other stakeholders regarding the factors that contribute to the autonomous vehicle crashes and guide the development of safer and more efficient autonomous vehicle systems.

The paper begins with a brief but comprehensive literature review that is followed by a description of the data preparation. Next, the methodology of the models' estimation is explained, and finally, the results and conclusions are presented.

2. LITERATURE REVIEW

The vehicles with self-driving capabilities have attracted a lot of attention among the researchers, engineers, and the public. Although the vehicles with automated driving have been marketed for their potential to reduce human driver costs, their effects on public transportation planning are still unclear as the public has been expressing concerns regarding their adoption and usage [3] [4] [5]. Former literature recognized safety as one of the most frequently identified concerns related to their acceptance [6] [7] [8]. Furthermore, a substantial part of past research focused on exploring the intended adoption of automated vehicles as well as identifying the essential barriers regarding their use.

To begin understanding crashes and their related injury severity that involves vehicles with varied levels of automation, several steps must be taken. Data availability and management are essential in attempting to gain relevant and applicable insights. Because of the data

availability, the studies addressing their actual real-life usage, related crashes, and users' experience have been scarce [9] [10]. Reliable data sources warrant accurate analysis and allow to plan for their effective and safe operation. A sizable amount of data is required to understand AV's safety performance. The California Department of Motor Vehicles is one of the most critical data sources that has been used thus far to investigate AV crashes. Former research by Novat et al. as well as Sinha et al. used these data to explore and identify crash patterns [11] [12]. Lee et al., on the other hand, used the reports from the National Highway Traffic Safety Administration (NHTSA) to examine automated vehicle crash patterns [2].

To more accurately identify potential risks and direct deployment plans, former research used a range of statistical models and applied them to analyze AV crashes. These methods included descriptive statistics and Bayesian inference [11] [13], data mining methods, such as machine learning models [14-16], classification and regression tree (CART) [17, 18], as well as deep neural networks that are capable of revealing patterns within observations. Depending on the available data sources, text mining has also been widely used to explore incident causality from crash reports [2, 19-21]. A summary of past research is shown in Table 1.

Former work that employed effective modeling and data pattern recognition was able to reveal several crash risk factors and provided insights for automated vehicle operations in practice. Past studies found that rear-end collision was an important crash type for vehicles with automated driving systems as well as different operation modes [22]. Ashraf et al. showed in their research that the majority of AV crashes were rear-end collisions while using the autonomous mode at road intersections [23]. Petrović et al. concluded that while the AV crashes involving conventional vehicles yielded more rear-end collisions, they also yielded fewer "pedestrian" or "broadside" collisions or aggressive driving behaviors [22]. Xu et al. performed

more systematic crash analysis using the interaction data from connected and autonomous vehicles and identified the main patterns of car crashes, which included the location type or driving mode [24].

Table 1. Summary of research related to the crashes of vehicles with driving automation.

Literature	Data	Methods
[2]	260 self-driving car crash reports	Path analysis framework and frequentist and Bayesian methods
[11]	127 self-driving car crashes and 865 conventional car crashes	Bayesian network (BN) model
[12]	259 crashes from CA DMV	Bagging
[13]	151 self-driving car crashes in California	Bayesian latent classification model
[15]	874 observations in AV disengagements and crashes	Nested logit model
[16]	131 crash reports	Apriori, SVM, CART, XGBoost
[17]	107 reports of traffic collisions involving self-driving cars	Ordered logistic regression and CART classification tree
[18]	245 trip AV collision records	The classification and regression tree (CART) model
[20]	124 traffic collision reports	Bayesian fixed and random parameters logit model
[23]	198 autonomous-vehicle-involved crash	Decision trees and association rules
[24]	72 crashes involving CAV	Bootstrap-based logistic regression method
[25]	252 self-driving cars crash data cases	Support vector machines (SVM), naïve Bayesian (NB), random forest (RF) and neural network (NN)
[26]	333 human-extracted data from self-driving car crashes reports	Bayesian network (BN) joint modeling method

Although some of the former studies have begun to examine the patterns and contributing factors to crashes related injury severity involving vehicles with automated driving systems, the research gap in this domain is still substantial. The knowledge regarding AV crashes remains fragmented, which stems primarily from the lack of data. Because of the data-related constraints,

there has been no comprehensive study on the injury severity that involves vehicles with driving automation capabilities. For example, limited research has been carried out to compare the crash patterns between the ADS and ADAS [27-29]. Most importantly, thus far the existing data sources have not been completed, which in consequence has created a risk for unreliable analysis and incomplete policy recommendations. The current analysis aims to deepen the understanding of factors impacting crash injury severity relating to vehicles with automated driving capabilities of different levels.

3. DATA PREPARATION

3.1. DATA COLLECTION

The current study uses worldwide crash data collected from the National Highway Traffic Safety Administration (NHTSA), the California Department of Motor Vehicles (CA DMV), and news reports. A total of 1,503 crash records of vehicles equipped with SAE Level 2 and above were collected. In addition, crashes reported in the news or on social media were also gathered to enrich the database. To differentiate between the vehicles with different automation levels, the vehicles were divided into two classes: automated driving systems (ADS) and advanced driver assistance systems (ADAS). The ADAS dataset corresponds to the crash records of the vehicles that are equipped with SAE Level 2 capabilities, while the ADS dataset includes information on crashes relating to the vehicles with the automation systems from SAE Levels 3 to Level 5. After cleaning the data and performing manual data quality control, a total of 709 and 571 crash records remained for the vehicles with the ADAS and ADS, respectively.

NHTSA crash database

In June 2021, NHTSA implemented a Standing General Order that requires manufacturers and operators to report crashes in detail for vehicles that are equipped with automated driving systems (SAE Levels 3 to 5) or advanced driver assistance systems (SAE Level 2). Hence, the NHTSA crash database includes the information on the ADAS and ADS

vehicle crashes from June 2021 that are collected from NHTSA's Crash Investigation Sampling System, Crash Report Supplement Datasheets, and Special Crash Investigations. Since crash records may be incomplete and unverified, NHTSA filtered and verified each crash record manually to prevent invalid, missing, or duplicated data.

CA DMV crash report

The other data source that was used in the study was the CA DMV OL316 Report of Traffic Collision Involving an Autonomous Vehicle. California DMV also requires the manufacturers that are testing the autonomous vehicles within the state of California to report any collision that resulted in property damage, bodily injury, or death within 10 days of the crash. The report form also includes the details that are related to the crash type, damage, crash location, and injury severity, as well as the crash scene information such as the weather, lighting condition, surface condition, pre-crash conditions, together with other associated factors often used for the AV crash analysis. The crash reports are available upon request, and the current analysis includes all the reported crashes from 2014 to March 2023 (the data were requested in March 2023 from the California Department of Motor Vehicles) [30].

Because the crash reports are provided in pdf format and their design often varies by year, it would be cumbersome to manually retrieve the key information from the original reports. To overcome this issue, we employed the pdf parser and an Large Language Model (LLM) such as ChatGPT to automatically extract crash attributes from the raw reports [31]. To ensure data quality and validity, we conducted manual checks and corrections for each report after the automated process was completed.

News reports

To enrich the data sources and samples, we also processed and collected the news and media posts about crashes of vehicles with SAE Level 2 automation or above. Since the news reports originated from a large social media community and required a substantial processing effort to obtain valuable and reliable data, we applied Large Language Model (LLM) to extract key information and align it with the NHTSA data format. Following the data processing, we also manually checked each file to eliminate any misedited or erroneous data points. The final sample included 35 valid crash records.

The three abovementioned data sources were further combined by matching the same or related attributes across the two sources and organizing them to align with the NHTSA crash database format. It must be noted that the duplicate crash records that were found in the NHTSA and CA DMV together with the redundant data were removed during the manual quality check. Furthermore, to contribute to the open-source research community, the processed and combined data were made available to the public and denoted as Autonomous Vehicle Operation Incident Dataset (AVOID). Detailed information about each step of data processing and gathering and their format is available in Zheng et al. [32].

3.2. DATA DESCRIPTION

To model the injury severity for vehicles with different automation levels, the dataset was divided into two categories. The first one included the information on the crashes of the vehicles equipped with the ADAS (that still require a driver), while the second one contained information on crashes of vehicles equipped with the ADS (capable of automated driving). The final datasets contain various independent variables that are categorized into four major classes: **road and environments characteristics** (including weather, road type, road description, surface condition, and lighting condition), **pre-crash characteristics** (including vehicle manufacture year, mileage,

driver type, speed, pre-crash movement, and crash impact area), crash impact area, and crash details characteristics (including crash severity, airbag engagement, and vehicle towing information). The injury severity levels were labeled into three categories: no injury, minor injury, and moderate/severe injury (including fatalities). A summary of the full sample of the available variables' characteristics is presented in Table 2.

A few interesting findings can be immediately noted by examining Table 2. For instance, for the ADAS equipped vehicles, over half of the crashes took place on a highway (56 percent), while most crashes involving the ADS equipped vehicles took place in more urban settings such as intersections (45 percent) and on local streets or roads (39 percent). Also, compared to the vehicles equipped with the ADAS, a higher percentage of crashes involving vehicles equipped with the ADS took place during better driving conditions (90 percent on dry surface and 94 percent in daylight), as shown in Figure 1.

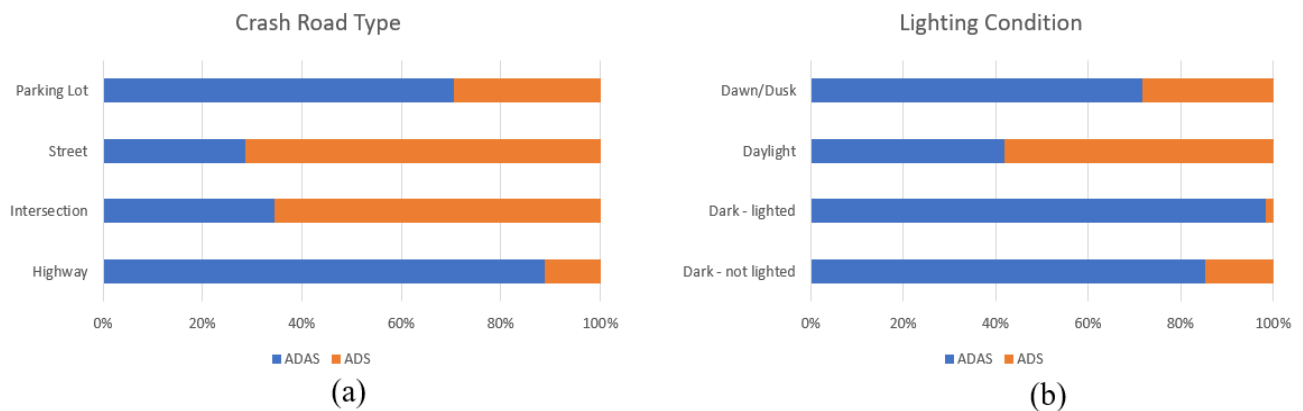


Figure 1. Distribution of crash types and lighting condition between the ADAS and ADS.

Furthermore, the crash speed shows obvious differences between the crashes involving the ADAS and ADS equipped vehicles. The speed for crashes involving the ADS equipped vehicles is found to be significantly higher compared to the speed of the ADAS equipped vehicles according to the results of the Chi-square test ($X^2=617.89$, $df=3$). Interestingly, there are

123 crashes involving the ADS equipped vehicles (Level 3-5) traveling the wrong way and consequently causing a collision, while there is no such cases for vehicles equipped with the ADAS.

Table 2. Summary statistics for the variables included in the generated datasets.

Variable type	Variable description	ADAS		ADS	
		Mean	Standard deviation	Mean	Standard deviation
Road and environment					
Weather	Clear sky indicator (1 if weather is clear, 0 otherwise)	0.78	0.17	0.77	0.26
	Cloudy sky indicator (1 if weather is cloudy, 0 otherwise)	0.07	0.07	0.13	0.08
		0.15			
	Rain indicator (1 if rain, 0 otherwise)		0.13	0.10	0.16
Road type	Highway indicator (1 if a highway, 0 otherwise)	0.56	0.25	0.09	0.33
	Intersection indicator (1 if an intersection, 0 otherwise)	0.19	0.16	0.46	0.20
	Street indicator (1 if road is a local street, 0 otherwise)	0.13	0.11	0.39	0.14
	Parking lot indicator (1 if a parking lot, 0 otherwise)	0.12	0.10	0.06	0.13
Road description	No unusual conditions (1 if there are no unusual conditions, 0 otherwise)	0.68	0.22	0.91	0.31
	Traffic incident/work zone (1 if traffic incident/work zone, 0 otherwise)	0.32	0.22	0.09	0.28
Surface	Dry surface indicator (1 if surface is dry, 0 otherwise)	0.54	0.25	0.91	0.33
	Wet surface indicator (1 if surface is wet, 0 otherwise)	0.31	0.21	0.08	0.27
	Snow/slush/ice indicator (1 if surface is covered with snow/slush/ice, 0 otherwise)	0.15	0.13	0.01	0.16
Lighting	Dark– not lighting (1 if it is dark– not lit), 0 otherwise)	0.04	0.04	0.01	0.05
	Dark– lighting (1 if it is dark– lit, 0 otherwise)	0.33	0.22	0.01	0.28
	Daylight (1 if daylight, 0 otherwise)	0.54	0.25	0.94	0.33
	Dawn/dusk indicator (1 if it is dawn/dusk, 0 otherwise)	0.09	0.08	0.040	0.10
Pre-crash conditions					
Manufacture year	Made before 2020 (1 if a vehicle was made before 2020, 0 otherwise)	0.47	0.25	0.60	0.33
	Made in or after 2020 (1 if a vehicle was made in and after 2020, 0 otherwise)	0.53	0.25	0.40	0.33
Mileage	Below or equal to 50,000(1 if a vehicle mileage is less than or equal to 50,000 miles, 0 otherwise)	0.68	0.22	0.86	0.31
	Above 50,000(1 if a vehicle mileage is more than 50,000 miles, 0 otherwise)	0.32	0.22	0.14	0.28
Driver type	Custom (1 if driver type is custom, 0 otherwise)	0.92	0.07	0.66	0.16
	Test (1 if driver type is test, 0 otherwise)	0.06	0.06	0.17	0.07
	Remote (1 if Driver type is remote, 0 otherwise)	0.02	0.01	0.17	0.01

Speed	Below or equal to -20 mph (1 if speed is below or equal to 20 mph, 0 otherwise)	0.14	0.12	0.83	0.15
	More than 20mph and less than 40 mph (1 if speed is more than 20 mph and less than 40 mph, 0 otherwise)	0.45	0.25	0.09	0.32
	More than 40 mph and less than 60 mph (1 if speed is more than 40 mph and less than 40 mph, 0 otherwise)	0.22	0.17	0.05	0.21
	Above or equal to 60 mph (1 if vehicle speed is above or equal to 60 mph, 0 otherwise)	0.19	0.14	0.02	0.18
Pre-crash movement	Proceeding straight (1 if pre-crash movement is proceeding straight, 0 otherwise)	0.68	0.22	0.44	0.31
	Merging (1 if pre-crash movement is merging, 0 otherwise)	0.12	0.11	0.01	0.13
	Crossing roadway (1 if pre-crash movement is crossing roadway, 0 otherwise)	0.05	0.05	0.12	0.06
	Turn (1 if pre-crash movement is turn, 0 otherwise)	0.03	0.03	0.13	0.04
	Changing lanes (1 if pre-crash movement is changing lanes, 0 otherwise)	0.05	0.05	0.07	0.06
Collision type	Stopping (1 if pre-crash movement is stopping, 0 otherwise)	0.06	0.06	0.02	0.07
	Fixed object (1 if crash with fixed object, 0 otherwise)	0.34	0.22	0.12	0.29
	Vehicle (1 if crash with another vehicle, 0 otherwise)	0.28	0.20	0.53	0.26
	Truck and van (1 if crash with truck and van, 0 otherwise)	0.27	0.20	0.21	0.02
	Motorcycle (1 if crash with motorcycle, 0 otherwise)	0.09	0.09	0.12	0.25
	Pedestrian (1 if crash with pedestrian, 0 otherwise)	0.02	0.02	0.02	0.11
AV Vehicle impact area					
	Left (1 if impact area is left, 0 otherwise)	0.14	0.12	0.15	0.15
	Right (1 if impact area is right, 0 otherwise)	0.11	0.1	0.18	0.12
	Rear (1 if impact area is rear, 0 otherwise)	0.4	0.24	0.47	0.31
	Front (1 if impact area is front, 0 otherwise)	0.12	0.11	0.16	0.13
	Top or bottom (1 if impact area is top or bottom, 0 otherwise)	0.04	0.04	0.05	0.05
Crash details					
Severity	No-injury (1 if injury severity outcome is no-injury, 0 otherwise)	0.80	0.16	0.89	0.25
	Minor (1 if injury severity outcome is minor, 0 otherwise)	0.14	0.12	0.09	0.15
	Moderate and higher (1 if injury severity outcome is moderate and higher including fatality, 0 otherwise)	0.06	0.06	0.02	0.07
Property damage	No damage (1 if there is no damage to a property, 0 otherwise)	0.03	0.03	0.08	0.03
	Damaged (1 if there is damage to a property, 0 otherwise)	0.97	0.03	0.92	0.11
Airbags	Engaged (1 if airbags are engaged, 0 otherwise)	0.88	0.11	0.28	0.19
	Not engaged (1 if airbags are not engaged, 0 otherwise)	0.12	0.10	0.72	0.13
Towed	Towed (1 if vehicle is towed, 0 otherwise)	0.08	0.07	0.22	0.09
	Not towed (1 if vehicle is not towed, 0 otherwise)	0.92	0.08	0.78	0.16

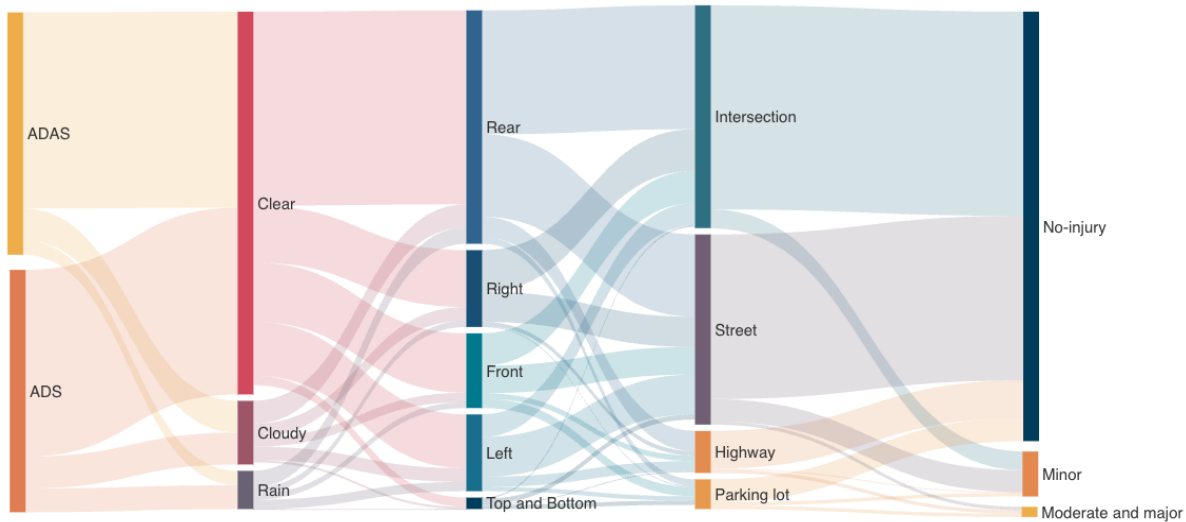


Figure 2. Relationship between the factors influencing injury severity.

Figure 2 presents the associations between the vehicle types, crash features, and injury severity levels. The width of each band corresponds to the proportion of the previous layer that contributes to the next layer. It can be concluded that most crashes took place in clear weather conditions, at intersections or streets, and impacted the rear of the vehicle.

4. METHODOLOGY

To gain more insights into the factors that play a role in the injury severity level, two random parameter multinomial logit models with heterogeneity in the means and variances were estimated. In each model, three injury severity categories were considered: no injury, minor injury, and moderate/severe injury (including fatalities). The chosen modeling approach allows the mean and variance of random parameters to be functions of explanatory variables and thus provides additional accuracy in studying unobserved heterogeneity. Random parameter ordered probit model was also estimated; however, it offered less interpretability and a worse fit. The multinomial logit model with heterogeneity in the means and variances has been widely applied

in traffic safety analysis [33] [34] . First, a function that determines injury severity probabilities is defined as,

$$F_{kn} = \beta_k \mathbf{X}_{kn} + \varepsilon_{kn}, \quad (1)$$

where F_{kn} is the function determining the probability of injury-severity category k in crash n , $\mathbf{X}_{kn}$ is the vector of explanatory variables that affect motorcyclist injury-severity level k , β_k is the vector of estimable parameters, and ε_{kn} is the error term which is assumed to be a generalized extreme value distributed [35]. Given that, a random parameter multinomial logit model of injury severity probabilities can be derived as [36]:

$$P_n(k) = \int \frac{\text{EXP}(\beta_k \mathbf{X}_{kn})}{\sum_{\forall K} \text{EXP}(\beta_k \mathbf{X}_{kn})} f(\beta_k | \boldsymbol{\varphi}_k) d\beta_k, \quad (2)$$

where $P_n(k)$ is the probability that crash n results in injury category k , $f(\beta_k | \boldsymbol{\varphi}_k)$ is the density function of β_k and $\boldsymbol{\varphi}_k$ refers to the vector of parameters describing the mixing density function (mean and variance), with all the other terms being previously defined. To give more flexibility in accounting for unobserved heterogeneity, with the mixing distribution now allowing parameters to vary across observations n , the β_{kn} vector can be made to be a function of variables that affect its mean and variance as [37] [38]:

$$\beta_{kn} = \beta_k + \Theta_{kn} \mathbf{Z}_{kn} + \sigma_{kn} \text{EXP}(\Psi_{kn} \mathbf{W}_{kn}) v_{kn} \quad (3)$$

where β_k is the mean parameter estimate across all crashes, $\mathbf{Z}_{kn}$ is a vector of observation-specific explanatory variables that captures heterogeneity in the mean that affects injury-severity level alternative k , Θ_{kn} is a corresponding vector of estimable parameters, $\mathbf{W}_{kn}$ is a vector of observation-specific explanatory variables that captures heterogeneity in the standard deviation (variance) σ_{kn} with corresponding parameter vector Ψ_{kn} and v_{kn} is a disturbance term.

The model estimation was performed by simulated maximum likelihood approach with 1,000 Halton draws as they were found to deliver more efficient distribution of simulation draws than purely random draws [33]. Although numerous density functions can be assumed for $f(\boldsymbol{\beta}_k|\boldsymbol{\varphi}_k)$ (Equation 2), normal distribution was found statistically superior therefore it will be used in the upcoming analysis, which is also consistent with past work [38] [39] .

To support the interpretation of the model estimation findings, marginal effects were calculated. Marginal effects capture the effect that a one-unit change in an explanatory variable has on the probability of an injury-severity outcome. The values of the corresponding marginal effects were calculated for each observation and were averaged over the population of observations. For variables with mixed effects, which are the variables with significant parameters in more than one injury severity function, net marginal effects are reported.

5. RESULTS

The estimation results of injury severity for crashes involving vehicles equipped with the ADAS (SAE Level 2, driver required) are presented and followed by the results for the ADS equipped vehicles (SAE Level 3 to Level 5, a driver is not required). Because these are two different levels of driving automation that likely exhibit different behavior in traffic, the likelihood ratio test was performed to ensure and validate the need for two separate models. The likelihood ratio test rejected the null hypothesis that both models are the same with over 95% confidence. Thus, two separate estimations are presented for injury severity outcomes for vehicles with different levels of automation (Table 3 and Table 4).

Estimating the random parameter models that are capable of capturing the heterogeneity in the means and variances of random parameters allows to gain detailed insights into the factors impacting the crash injury severity. To ensure that the random parameter models are indeed

statistically superior to the fixed parameter models, we performed the likelihood ratio test. The results confirmed that the random parameter models are superior at 99% confidence level. In consequence, the random parameter models serve as the foundation for the following discussions and findings in this study.

Table 3. Random parameter logit model with heterogeneity in the mean and variances of random parameters on crash-injury severity for vehicles equipped with ADAS (SAE Level 2) (parameters defined for [N] No injury; [M] Minor injury; [MM] Moderate/severe injury).

Variable	Estimated parameter	t-statistic	Marginal effects		
			No injury	Minor injury	Moderate/severe injury
Constant [M]	-2.37	-2.16			
Constant [MM]	-1.78	-1.09			
<i>Random parameters in utility functions</i>					
Below or equal to 50,000(1 if a vehicle mileage is less than or equal to 50,000 miles, 0 otherwise) [M]	-7.10	-2.05	-0.0159	-0.0031	0.0189
Standard deviation of parameter distribution	5.09	2.42			
<i>Heterogeneity in the mean of the random parameter</i>					
Mileage below or equal to 50,000 [M]: Wet surface indicator (1 if surface is wet, 0 otherwise)	3.52	2.05			
<i>Heterogeneity in the variance of the random parameter</i>					
Mileage below or equal to 50,000 miles [M]: Front impact (1 if vehicle impact area is front, 0 otherwise)	0.29	1.69			
<i>Road and environment</i>					
Rainy weather indicator (1 if rain, 0 otherwise) [M]	-2.202	-4.27	0.00206	-0.00249	0.00033
Wet surface indicator (1 if surface is wet, 0 otherwise) [MM]	1.57	4.1	-0.056	-0.0047	0.0607
Intersection indicator (1 if an intersection, 0 otherwise) [M]	2.37	3.03	-0.0051	0.0069	-0.0017
<i>Crash details</i>					
Airbag engagement indicator (1 if airbags were engaged, 0 otherwise) [MM]	-4.51	-9.54	0.2456	0.017	-0.2627
<i>Pre-crash conditions</i>					
Driver type indicator (1 if driver type is custom, 0 otherwise) [M]	4.21	1.88	-0.0032	0.0040	-0.0008
Driver type indicator (1 if driver type is test, 0 otherwise) [MM]	-3.26	-3.8	0.0156	0.0004	-0.0160
Vehicle medium speed indicator (1 if speed is more than 20mph and less than 40 mph, 0 otherwise) [M]	-1.78	-1.86	0.0049	-0.0054	0.0004
Older vehicle indicator (1 if a vehicle was made before 2020, 0 otherwise) [MM]	1.22	3.81	-0.0444	-0.0038	0.0482
Mileage above 50,000 miles (1 if a vehicle	2.5	3.62	-0.0102	-0.0009	0.0110

mileage is more than 50,000 miles, 0 otherwise) [MM]					
Variables with mixed effects (variables with significant parameters in more than one injury severity function, net marginal effects reported)					
Collision with fixed object indicator (1 if crash with fixed object, 0 otherwise) [M]	-3.74	-4.02	0.0386	-0.0095	-0.0291
Collision with fixed object indicator (1 if crash with fixed object, 0 otherwise) [MM]	-1.25	-3.34			
Rear impact (1 if vehicle impact area is rear, 0 otherwise) [M]	-1.5	-2.93	-0.0493	-0.0164	0.0657
Rear impact (1 if vehicle impact area is rear, 0 otherwise) [MM]	1.59	3.43			
Front impact (1 if vehicle impact area is front, 0 otherwise) [M]	-1.58	-3.81	-0.013	-0.0214	0.0344
Front impact (1 if vehicle impact area is front, 0 otherwise) [MM]	1.18	2.89			
Number of observations			709		
Log likelihood at zero, LL(0)			-314.42		
Log likelihood at convergence, LL(β)			-778.92		
$\rho^2 = 1 - LL(\beta)/LL(0)$			0.60		

Table 4. Random parameter logit model with heterogeneity in the mean of random parameter on crash-injury severity for vehicles equipped with ADS (SAE Level 3 to Level 5) (parameters defined for [N] No injury; [M] Minor injury; [MM] Moderate/severe injury).

Variable	Estimated parameter	t-statistic	Marginal effects		
			No injury	Minor injury	Moderate/severe injury
Constant [M]	-2.81	-2.64			
Constant [MM]	-2.12	-1.92			
Random parameters in utility functions					
Mileage below or equal to 50,000 miles (1 if a vehicle mileage is below or equal to 50,000 miles, 0 otherwise) [M]	-4.20	-3.98	0.0099	-0.0131	0.0032
Standard deviation of parameter distribution	3.35	2.75			
Heterogeneity in the mean of the random parameter					
Mileage below or equal to 50,000 miles [M]: Rear (1 if impact area is rear, 0 otherwise)	2.72	1.62			
Road and environment					
Rainy weather indicator (1 if rain, 0 otherwise) [M]	-2.59	-2.53	0.0078	-0.0089	0.0011
Cloudy sky indicator (1 if the sky is cloudy, 0 otherwise) [MM]	-1.84	-2.23	0.0070	0.0019	-0.0089
Intersection indicator (1 if an intersection, 0 otherwise) [M]	-1.32	-2.92	0.0197	-0.0210	0.0013
Crash details					
No airbag engagement indicator (1 if airbags were not engaged, 0 otherwise) [MM]	5.07	4.56	-0.0273	-0.0150	0.0423

<i>Pre-crash conditions</i>					
Driver type indicator (1 if driver type is test, 0 otherwise) [M]	4.11	7.50	-0.0751	0.0876	-0.0125
Pre-crash movement indicator (1 if pre-crash movement is turn, 0 otherwise) [M]	-1.62	-2.14	0.0053	-0.0059	0.0006
Pre-crash movement (1 if pre-crash movement is stopping, 0 otherwise) [M]	-3.13	-3.29	0.0481	-0.0549	0.0068
Vehicle medium speed indicator (1 if speed is more than 20 mph and less than 40 mph, 0 otherwise) [M]	-1.65	-3.43	0.0057	-0.0124	0.0067
Vehicle high speed indicator (1 if vehicle speed is above 60 mph, 0 otherwise) [MM]	5.98	2.95	-0.0089	-0.0040	0.0129
Collision with fixed object indicator (1 if crash with fixed object, 0 otherwise) [MM]	-3.59	-4.49	0.0162	0.0073	-0.0235
Mileage is below or equal to 50,000 miles (1 if a vehicle mileage is below or equal to 50,000 miles, 0 otherwise) [MM]	-5.21	-2.76	0.0074	0.0040	-0.0114
<i>Variables with mixed effects (variables with significant parameters in more than one injury severity function, net marginal effects reported)</i>					
Wet surface indicator (1 if surface is wet, 0 otherwise) [M]	3.41	4.28	-0.0274	0.0220	0.0054
Wet surface indicator (1 if surface is wet, 0 otherwise) [MM]	3.32	3.18			
Parking lot indicator (1 if a parking lot, 0 otherwise) [M]	-1.32	-2.73	0.0395	-0.0195	-0.0200
Parking lot indicator (1 if a parking lot, 0 otherwise) [MM]	-5.19	-4.47			
Traffic incident/work zone (1 if traffic incident/work zone, 0 otherwise) [M]	3.55	6.09	-0.0393	0.0242	0.0151
Traffic incident/work zone (1 if traffic incident/work zone, 0 otherwise) [MM]	4.22	4.77			
Rear impact (1 if vehicle impact area is rear, 0 otherwise) [M]	-1.05	-2.22	0.0315	-0.0222	-0.0093
Rear impact (1 if vehicle impact area is rear, 0 otherwise) [MM]	-3.92	-3.32			
Newer vehicle indicator (1 if a vehicle was made in or after 2020, 0 otherwise) [M]	-1.70	-3.66	0.0431	-0.0239	-0.0192
Newer vehicle indicator (1 if a vehicle was made in or after 2020, 0 otherwise) [MM]	-9.19	-4.90			
Number of observations			571		
Log likelihood at zero, LL(0)			-627.31		
Log likelihood at convergence, LL(β)			-113.76		
ρ ² = 1–LL(β)/LL(0)			0.82		

Tables 3 and 4 present model estimation results of the random parameter multinomial logit models with heterogeneity in the means and variances on the crash injury severity for vehicles equipped with the ADAS and ADS, respectively. A total of 13 variables were found significant in the model examining crash injury severity involving the ADAS (SAE Level 2)

equipped vehicles (Table 3), and a total of 17 variables were found significant in the model analyzing crash injury severity involving ADS (SAE Level 3-5) equipped vehicles (Table 4).

5.1 SELECT FINDINGS ON THE ROAD AND ENVIRONMENT FACTORS AND CRASH DETAILS

Rainy weather was found to decrease the probability of a minor injury for both ADAS and ADS equipped vehicles relative to other weather conditions. Although historically, rainy conditions have been found hazardous for driving, some drivers were found to be more cautious and maintain large gaps in adverse weather conditions [40] (which would be particularly relevant for the ADAS vehicles). Furthermore, considering the crashes involving vehicles with ADAS (SAE Level 2), automatic emergency braking systems can detect an impending collision and apply the brakes to prevent or mitigate the impact. In rainy conditions, these systems can detect the need to brake earlier and with less force. Secondly, although rain can increase the chances of a vehicle skidding or losing control, the ADAS technology such Automatic Emergency Braking (AEB), and Adaptive Cruise Control (ACC) can help prevent or correct skids and loss of control [41]. Regarding the vehicles with higher levels of automation, since their technology relies on pre-programmed algorithms and sensors, ADS may also react more cautiously to these weather conditions [42]. Because both models captured the importance of the rain indicator, it would be important to include adverse weather conditions in any future studies related to the crash injury severity that involve vehicles with automated driving systems.

Interestingly, the wet surface indicator indicates a 0.0607 higher probability of moderate/severe injury for vehicles equipped with the ADAS (as indicated by the marginal effects in Table 3). The performance of sensors and cameras, which are vital components of the ADAS technology, can be affected by wet surfaces. Wet surface indicator results also increase

the probability of minor and moderate/severe injuries for the ADS equipped vehicles (as indicated by the net marginal effects). Moisture or raindrops on the camera lens can cause distortion or blockage of the view, resulting in inaccurate or incomplete data inputs [43]. In consequence, it can lead to errors in the decision-making process of the ADAS equipped vehicle, resulting in delayed responses or incorrect reactions.

Concerning the road type, crashes occurring at the intersections result in a higher probability of minor injuries for the vehicles equipped with the ADAS (Table 3) and decrease the probability of minor injuries for the ADS equipped vehicles (as indicated by the marginal effects presented in Table 4). This finding indicates that the injury severity outcome will likely vary by the level of automation and environment. Former research argues that the ADAS equipped vehicles may encounter difficulties in detecting other vehicles, pedestrians, or hazards at intersections with poor visibility, and limited sensor range in low visibility conditions can result in failure to detect a vehicle or crossing pedestrian [44]. Furthermore, turning or merging vehicles are frequently present at intersections, which can be challenging for the ADAS equipped cars to detect and respond swiftly [44]. Although more research is needed to test the existing systems, the current analysis points to the variation in injury severity between the ADAS and ADS at intersections, which can be explained by the degree of control and the intervention of the system. The vehicles equipped with the ADAS still depend on human drivers to maneuver intersections, while the ADS equipped vehicles are engineered to function autonomously, relying on sensors and algorithms to detect and react to the surroundings [44]. Furthermore, the fact that the operator is aware of these systems can change their behavior and their level of alertness.

Turning to the crash location, while the parking lot indicator results in lower probabilities of minor and moderate/severe injuries for the ADS equipped vehicles (as indicated by the net

marginal effects), traffic incident/work zone indicator results in an increased probability of minor and moderate/severe injuries (as suggested by the net marginal effects in Table 4). Traffic work zones represent an area of a roadway where construction, maintenance, or utility work activities are identified by warning signs, signals, or other indicators, and traffic incident indicates a prior traffic event that adversely affects normal traffic operations separate from the reported incident. Both of them can bring about unanticipated and abrupt modifications to driving conditions, detours, lane closures, or reduced speed limits [45]. These alterations can pose a challenge for the ADS to perceive and react to, increasing the possibility of a crash. Additionally, work zones and traffic incidents usually entail a greater concentration of vehicles, pedestrians, and equipment, creating more obstacles for the ADS to identify and evade.

The impact of airbags engagement has been found particularly remarkable. Airbag engagement indicator results in a 0.2627 lower probability of moderate/severe injury for vehicles equipped with SAE Level 2 (driver required). The model estimation results for the vehicles equipped with ADS, on the other hand, offer an interesting finding. Lack of airbag engagement indicator results in a 0.0423 higher probability of a moderate/severe injury. Although the finding is to some degree intuitive, it could also signal that reconsidering the circumstances under which the airbags get engaged could be beneficial to decreasing the crash injury severity outcomes. Because the crash happened and the airbags did not get deployed, the manufacturers have the opportunity to examine the crash characteristics and improve the vehicle safety.

5.2 SELECT FINDINGS ON THE INFLUENCE OF CRASH DETAILS AND THE PRE-CRASH CONDITIONS

It is interesting to find that the drive type indicator was found statistically significant in both models and increased the probability of minor injuries. In case of the vehicles equipped

with ADAS, the custom drive type indicator resulted in a 0.004 higher probability of a minor injury, whereas in the model developed for the ADS, the marginal effect for test drive type indicator was found much larger and increasing the probability of a minor injury by 0.0876 (as indicated by the marginal effects in Table 4). Remarkably, the test drive type indicator in the ADAS model was found to decrease the probability of a moderate/serious injury by 0.0160 as suggested by the marginal effects in Table 3. Studying the outcomes of injury severity for different drive type indicators and different levels of automation allows to further improve and optimize the operations and safety. Because the testing vehicles used for evaluating the ADAS are usually equipped with the latest and most advanced sensors, software, and hardware available and they also receive frequent updates and calibration, which can improve their performance and accuracy, it is particularly important to evaluate new technological capabilities in the context of injury severity outcomes. Furthermore, testing vehicles often have an experienced operator who is very familiar with the technology and can react in real time to any unsafe conditions [46] [47] [48].

Turning to the pre-crash condition such as movement and speed, the findings related to the ADS equipped vehicles reveal that both pre-crashes stopping or turning indicators result in a lower probability of a minor injury. It is likely because stopping and turning often involve more deliberate and regulated vehicle movements, these indicators were consequently found statistically significant in the model [47]. Besides, the reduced speeds and lower probability of certain types of collisions during these maneuvers can also contribute to a lower risk of minor injuries for the ADS equipped vehicles.

Speed related findings for the ADS equipped vehicles are worth noting. While the median speed variable indicating that the speed was more than 20 mph and less than 40 mph decreased

the probability of a minor injury, the median speed of at least 60 mph indicator increased the probability of a moderate/severe injury. While speed has been a known factor determining the injury severity, its impact may be different for the vehicles with higher level of automation. Highly automated vehicles can adjust their distance and react faster than human driver in many conditions and while speed is often considered endogenous by past studies [49, 50], it is important to investigate the role it plays in crash injury severity outcomes for vehicles with automation. Exploring the injury severity outcomes at different speeds first is essential to deepening the knowledge regarding the safety of autonomous vehicles. Second, it can be concluded that even the vehicles with high levels of automation (SAE Level 3 to Level 5) will not completely prevent severe injury outcomes for crashes at higher speeds as they have not been able to avoid them.

When turning to the collision object, it was found in both models that, a collision with fixed object resulted in a 0.0291 lower probability of a moderate/severe injury for the ADAS equipped vehicles (as indicated by the net marginal effects in Table 3) as well as a 0.0235 lower probability of a moderate/severe injury for the ADS equipped vehicles (Table 4). In cases of crashes with fixed objects, it could be argued that both, the ADAS and ADS, can detect obstacles ahead using sensors and cameras [43]. The vehicle can consequently take preventive measures such as applying brakes or other collision avoidance techniques to decelerate before the impact. This reduction in speed leads to a decrease in the impact force, thus lowering the injury severity. Additionally, both technologies can modify the vehicle's trajectory to minimize the impact force by ensuring that the collision occurs at an angle that reduces its force.

The indicator of mileage below 50,000 miles for the ADAS equipped vehicles was also found to produce statistically significant findings in the model estimation, but there was a

significant heterogeneity captured by this variable as indicated by the statistically significant random parameter. Furthermore, Table 3 shows that the vehicles with mileage below 50,000 miles that were engaged in a crash on a wet surface had an increase in their mean as well as an increase in the variance (as showed by the statistically significant variance noted as the front impact indicator). While the mileage indicator likely captures the overall wear and tear of the vehicle and its sensors, it is also interesting to note that there is a substantial heterogeneity in this variable, which creates an important opportunity for future research. It was also found that the older ADAS equipped vehicle indicator (vehicle manufactured before 2020) as well as high vehicle mileage indicator (mileage above 50,000 miles) both increase the probability of a moderate/severe injury by 0.0482 and 0.0110, respectively (as indicated by the marginal effects in Table 3). For the ADS equipped vehicles, it was discovered that a newer vehicle indicator (manufactured in or after 2020) resulted in a 0.0114 lower probability of moderate/severe injury. These findings indicate that as the technology progresses and the newer sensors become available on the market, they will likely offer further safety benefits. Like the ADAS vehicles, the indicator of mileage below 50,000 miles in the ADS model was also found to produce statistically significant findings in the model estimation and there was a significant heterogeneity captured by this variable as indicated by the statistically significant random parameter. Furthermore, Table 4 shows that those vehicles with mileage below 50,000 that were engaged in a crash that resulted in rear impacts (AV vehicle was rear ended) had an increase in their mean, indicating the importance of rear crashes and their role in the injury severity outcomes.

The location of impact has shown to be particularly complex as the indicators were found to be significant parameters in more than one injury severity function, which led to reporting the net marginal effects in both models (Table 3 and Table 4). It was found that the rear and front

impact indicators for the model using the crashes of vehicles equipped with the ADAS result in a higher probability of a moderate/severe injury by 0.0657 and 0.0344, respectively. On the contrary, the rear impact indicator was found to decrease the probability of minor and moderate/severe injuries in crashes of the ADS equipped vehicles. Although there could be numerous reasons for this phenomenon, it should be noted that while the vehicles equipped with the ADAS still require a driver, the ADS vehicles are capable of more automated driving and, therefore could offer more benefits in mitigating injury severity of rear impact crashes. ADS relies on a combination of sensors, algorithms, and real-time decision-making processes to navigate and operate a vehicle [36]. Rear-end crashes often occur at higher speeds (66.75 percent of rear-end crashes occurred on highways in the current dataset), leaving the drivers with little time to react, even with the help of ADAS. This can lead to more significant impacts and more severe injuries. Secondly, in some cases, the ADAS system may cause a rear-end collision. For example, a sudden and unexpected braking by the ADAS equipped vehicles can cause a following vehicle to collide with its rear.

6. SUMMARY AND CONCLUSIONS

This study conducted a thorough analysis using a multi-source dataset (for more details regarding the dataset please refer to Zheng et al. [30]), which consists of collision reports and crash news for vehicles of different levels of automation (SAE Level 2 to Level 5). To ensure the reliability of the analysis, 709 and 571 crash records were collected and reflected the crash details for vehicles equipped with the ADAS and ADS, respectively. Two random parameters logit models with heterogeneity in the means and variances were estimated to compare the ADS and ADAS collision data and crash injury severity outcomes. Four types of variables were considered, and they included the impact area, road and environment, pre-crash movement, and

crash details. Given the available variables, three levels of injury severity for the model estimation were used (no-injury, minor injury, and moderate/severe injury, including fatalities).

From the descriptive analysis it was found that over half of the crashes (56 percent), involving the ADAS equipped vehicles (SAE Level 2, driver required) took place on a highway, while most crashes involving the ADS equipped vehicles (SAE Level 3-5, no driver required) took place in more urban settings such as intersections (45 percent) and on local street or roads (39 percent). Interestingly, there were 123 crashes involving ADS equipped vehicles (Level 3-5) traveling the wrong way and consequently causing a collision, while there was no such a case for the vehicles equipped with the ADAS.

Turing to the model estimation results, the wet surface indicator was particularly important and resulted in a 0.0607 higher probability of a moderate/severe injury for vehicles equipped with the ADAS. Because moisture or raindrops on the camera lens can distort the view, these conditions can consequently lead to an increase in moderate/severe injuries on wet surfaces [43]. For instance, some rain sensors can facilitate rain intensity detection to optimize the speed of the windshield wipers [51]. Also, traction control can be applied to prevent wheel slippage on wet roads, which ultimately enhances the safety of ADS and ADAS [52]. Third, an integrative design with multiple sensors can also equip the ADAS and ADS vehicles to improve their safety. Lastly, combining multiple sources, including cameras, radar, and LIDAR, can provide diverse sensory input to ensure reliable and safety responses even when some sensors fail [53].

Considering crash location, traffic incident/work zone indicator resulted in an increased probability of minor and moderate/severe injuries that involve the ADS equipped vehicles (SAE Level 3-5) (as suggested by the net marginal effects in Table 4). Because traffic work zones and incident zones can lead to sudden unanticipated and abrupt modifications to the driving

environment, special attention has to be paid to the performance of the ADS equipped vehicles in work zones [45]. While the performance of the vehicles with automated driving systems in work zones has been identified to be particularly important in crash injury severity, a collision with fixed object resulted in a 0.0291 lower probability of a moderate/severe injury for the ADAS equipped vehicles (as indicated by the net marginal effects in Table 3) as well as a 0.0235 lower probability of a moderate/severe injury for the ADS equipped vehicles (Table 4). In cases of crashes with fixed objects, it was found that both the ADAS and ADS lower the probability of a moderate/severe injury during collisions with fixed objects [43].

Differences in the system sophistication that were captured by both— manufacture year and high mileage played a significant role in the crash injury severity. It was found that the older ADAS equipped vehicle indicator (vehicle manufactured before 2020) as well as high vehicle mileage indicator (mileage above 50,000 miles) both increase the probability of a moderate/severe injury (Table 3). Interestingly, the ADS equipped vehicles that were manufactured in or after 2020 were found to decrease the probability of a moderate/severe injury. These findings indicate that relatively small updates to the technology can result in significant shifts in the injury severity outcomes. Lastly, the location of impact has been especially important in this analysis as it was found that the rear and front impact indicators for the model using the crashes of vehicles equipped with the ADAS resulted in a higher probability of a moderate/severe injury. On the contrary, the rear impact indicator was found to decrease the probability of minor and moderate/severe injuries in crashes of the ADS equipped vehicles. These findings suggest that designing vehicles with automated capabilities should continue to account for human drivers in the fleet as the sudden and unexpected braking done particularly by the ADAS equipped vehicles can cause a following vehicle to collide with its rear. To reduce the

injury severity, manufactures may consider Automatic Emergency Braking (AEB), Blind Spot Monitoring (BSM), Lane Departure Warning (LDW), or Rear Cross-Traffic Alert (RCTA) [54]. Secondly, the sensors like LIDAR and cameras can be integrated to detect road condition and enhance the performance of ADS and ADAS technologies [55].

Although significant insights into the factors affecting the crash injury severity of vehicles capable of automated driving have been obtained by analyzing real-world and multi-source data, the current research has its limitations. One of the main drawbacks of this study is the limited number of observations indicating higher levels of crash injury severity. An alternative approach would be to use a binary approach (injury and no injury) or nested logit model however, some important pieces of information would be lost. Because it is an exploratory study, it was valuable to include all crash injury severities. Lastly, the scope of the study does not include performance comparison between the human-driven vehicles as the focus is primarily on the vehicles with SAE Levels 2-5. As the data become available, it is critical to examine the true benefits of automated vehicles and compare their results with human-driven vehicles in the real-world conditions.

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